

Ergon Energy Network Tariff Guide

1 July 2020 to 30 June 2021



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1. Introduction

1.1 Purpose

This document is Ergon Energy's Network Tariff Guide (Guide). It supports Ergon Energy's 2020-21 Pricing Proposal and has been prepared to assist in the interpretation of our network tariffs and tariff assignment processes for the period from 1 July 2020 to 30 June 2021.

Each year we are required to submit a pricing proposal to the the Australian Energy Regulator (AER) for approval. The pricing proposal sets out Ergon Energy's proposed tariffs and demonstrates compliance with Chapter 6 of the National Electricity Rules. Our 2020-21 Pricing Proposal was developed in accordance with the requirement set out in our 2020-25 Tariff Structure Statement (TSS).

This Guide aligns with our AER approved 2020-25 TSS and 2020-21 Pricing Proposal. Ergon Energy's Tariff Structure Statement, Pricing Proposal and Network Tariff Tables are available on our website: www.ergon.com.au/network/network-management/network-pricing.

1.2 Supporting network pricing documentation

In addition to this Guide, we have published a number of related network pricing documents to assist network users, retailers and interested parties understand the development and application of tariffs and connection charges.¹ These documents are outlined in the table below.

Table 1: Supporting network pricing documentation

Document	Overview
Tariff Structure Statement	- Sets out the proposed tariff classes, tariffs and tariff structures for the 2020-25 period - Details how the proposed tariff classes, tariffs and tariff structures comply with the pricing principles - Provides details on Ergon Energy's tariff assignment policy - Provides indicative prices for the 2020-25 regulatory control period - Approved by the AER as part of the 2020-25 Distribution Determination
Pricing Proposal	- Explains Ergon Energy's tariff classes, tariffs and tariff structures for Standard Control Services and Alternative Control Services in compliance with the requirements set out in Chapter 6 of the NER, the AER's Distribution Determination and our TSS - Submitted to the AER annually for approval
Network Tariff Schedule	- Provides Ergon Energy's prices for our Standard Control Services and Alternative Control Services developed in accordance with the requirements set out in the NER, the AER's Distribution Determination and our TSS - Submitted to the AER annually as part of the Pricing Proposal
Network Tariff Guide	- An operational document for customers, retailers, and consultants, setting out the tariff assignment and reassignment procedure - Provides a description of the network tariffs - Provides an explanation of the application of network tariff charging components - Published annually and updated as required
Connection Policy	- Sets out when a connection charge may be payable by retail customers or real estate developers and the aspects of the connection service for which a charge may be applied - Details how Ergon Energy calculates the capital contributions to be paid - Approved by the AER in 2020 as part of the 2020-25 Distribution Determination

¹ Link to Ergon Energy's website: www.ergon.com.au/network/network-management/network-pricing

1.3 Background

1.3.1 Network tariff charging components

The total network charges customers are charged for their use of the distribution network (i.e. for Standard Control Services) are known as Network Use of System (NUOS) charges.

NUOS charges are comprised of the following three components:

- Distribution Use of System (DUOS) charge – this charge refers to the network charge attributable to the use of Ergon Energy's distribution network.
- Designated Pricing Proposal Charge (DPPC) – this charge mainly refers to the charges incurred for the use of Powerlink's transmission network. It was previously referred to as Transmission Use of System (TUOS) charge.
- Jurisdictional scheme charges – this charge refers to the amounts imposed on Ergon Energy through legislative obligations by the Queensland Government.

2. Pricing zones

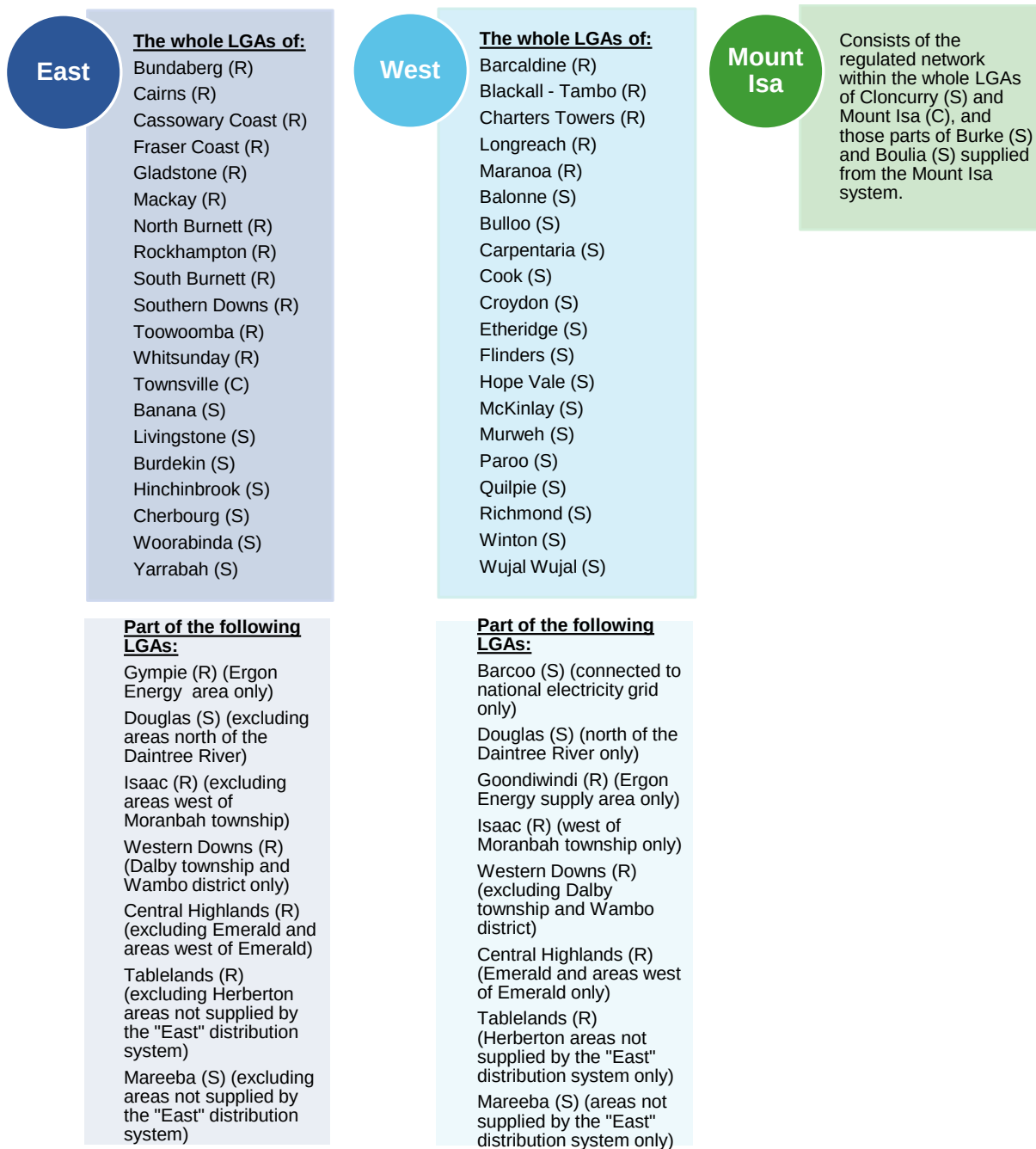
Three pricing zones have been delineated in our distribution area, broadly based on Queensland's local government areas, with the distribution network electrical connection being the final determinant of which zone applies. Zone pricing impacts the DUOS component of the NUOS charges only (TUOS charges and jurisdictional scheme charges are not impacted by pricing zones).

Ergon Energy's three pricing zones are:

- East Zone – those areas where the network users are supplied from the distribution system connected to the national grid and have a relatively low distribution cost to supply.
- West Zone – those areas outside the East Zone and connected to the national grid, which have a significantly higher distribution cost of supply than the East Zone.
- Mount Isa Zone – broadly defined as those areas supplied from the isolated Mount Isa system. This zone is not connected to the national grid and, as such, would normally be excluded from the application of the NER. However, under the Electricity – National Scheme (Queensland) Act 1997, the Queensland Government has transferred responsibility for the economic regulation of the Mount Isa – Cloncurry supply network to the AER.

The local government areas covered in each of the three regions are listed in the figure below.

Figure 1: Pricing zones coverage



Note: (R) = Regional Council, (S) = Shire Council and (C) = City Council

3. Assigning and reassigning customers to network tariff classes and tariffs

This chapter sets out Ergon Energy's procedures for assigning new customers² to a default network tariff and for reassigning existing customers to alternative NUOS tariff. This chapter should be read in conjunction with our approved 2020-25 TSS and the AER's 2020-25 TSS Decision.

New customer assignment and existing customer reassignment to Ergon Energy's default network tariff involves two steps:

- 1) assigning new customers or reassigning existing customers to the applicable tariff class based on their connection characteristics, and
- 2) assigning new customers or reassigning existing customers to the applicable network tariff within their correct tariff class.

3.1 Assigning new customers

3.1.1 Assignment to tariff class

Consistent with our TSS, Ergon Energy will assign customers into one of three tariff classes, mainly based on the voltage level at which customers are connected to the network. Ergon Energy's tariff classes and eligibility criteria are explained in Table 2: Tariff classes.

Table 2: Tariff classes

Tariff class	Eligibility criteria
Standard Asset Customers (SAC)	All customers connected at LV with installed capacity up to 1,000kVA are assigned to the SAC tariff class. SAC customers are further classified as Small or Large customers, depending on their energy consumption: • SAC Small – A small customer is defined in the National Energy Retail Law (Queensland) Act 2014 as a residential or small business customer with annual energy consumption up to 100 MWh. • SAC Large – A large customer is defined as an LV customer with annual energy consumption greater than that of a small customer as determined in Section 7 of the <i>National Energy Retail Regulations</i>, that is customers with annual energy consumption of 100 MWh or more.
Connection Asset Customers (CAC) ^a	Customers with a network coupling point at 66 kV, 33 kV, 22 kV, 11 kV and installed capacity above 1,000 kVA who are not assigned to the ICC tariff class are allocated to the CAC tariff class.
Individually Calculated Customers (ICC)	Customers are assigned to the ICC tariff class if they are coupled to the network at 132 kV, 110 kV, 66 kV or 33 kV, and with installed capacity above 10 MVA. Customers may also be assigned to the ICC tariff class if they are coupled to the network at 132 kV, 110 kV, 66kV or 33 kV, and with installed capacity below 10 MVA where: • A customer has a dedicated distribution system which is quite different and separate from the remainder of our distribution system • At the determination of the DNSP, the nature of the customer's connection to the network, and/or usage of the network, make average prices inappropriate • A customer is connected at or close to a Transmission Connection Point, or • Inequitable treatment of other customers would arise from the application of the 10MVA threshold. • Subject to the Policy set out in our TSS Appendix A, eligible CAC customers accessing transitional or obsolete retail tariffs and who can demonstrate that they are facing

² In this Guide, a new customer means as a new connection to the distribution network.

Tariff class	Eligibility criteria
	extraordinary customer impact post retirement of the retail tariffs and that this financial impact is directly attributable to their network charges.
Note: Some existing customers coupled to the HV network at lower voltage levels will remain allocated to the ICC tariff class for legacy reasons.	

No reference is made to customer's export load in assigning customers to Ergon Energy's tariff classes (or network tariffs).

3.1.2 Assignment to default primary tariff

If a retailer does not specify its preferred network tariff for a new customer, Ergon Energy will assign the customer to the default the network tariff in accordance with the table below. It should be noted that all new customers are assumed to have smart meters.

Table 3: Default tariff assignment for new customers

Tariff class	Customer type	Usage	Default network tariff	Tariff code
SAC	Residential	Below 100 MWh per annum	Residential Transitional Demand	RTDEM
	Small Business	Below 100 MWh per annum	Small Business Transitional Demand	BTDEM
	Residential or Business	100 MWh or more, and demand above 30 kW (or 35 kVA)	Demand Small	DST
	Residential or Business	100 MWh or more, and demand above 120 kW (or 135 kVA)	Demand Medium	DMT
	Residential or Business	100 MWh or more, and demand above 400 kW (or 450 kVA)	Large Business Time-of-Use Demand	LTOUD
	Unmetered supply	All	Unmetered supply	UM
CAC	Network coupling point at 11kV feeder shared with other customers		22/11 kV Line	C22L
	Network coupling point at an 11kV zone substation bus via a dedicated 11kV feeder		22/11kV Bus	C22B
	Network coupling point at 66 kV		66 kV	C66
	Network coupling point at 33 kV		33 kV	C33
ICC	All ICC customers		ICC tariff	ICC

Ergon Energy's network tariffs do not support a mixed tariff situation (for example, where one NMI has both residential and business retail tariffs). The determination of the appropriate SAC network tariff will be based on the retailer's classification of the NMI as either business or residential in accordance with the National Energy Retail Rules.

If a customer classification is not received from the retailer for move-in SAC small customers, the retail customer moving-in to the existing premises will inherit the existing customer classification and existing network tariff. Move-in customers are not considered as a new customer to Ergon Energy, as these customers are not a new connection to the distribution network.

Where a retailer has not nominated a valid network tariff for a SAC Large customer, Ergon Energy will use the available site information and attributes to estimate the load characteristics for the

premise in terms of Demand Small, Demand Medium or Large Business Time-of-use Demand, and on a best endeavours basis match those characteristics to the optimal default tariff.

3.2 Reassigning existing customers

In accordance with our TSS, Ergon Energy will initiate network tariff reassignment of customers in the following instances:

- when SAC customers change from a basic accumulation meter to a smart (Type 4) meter,
- when a SAC customer reaches the end of any applicable grace provision within the TSS,
- to transition SAC customers that already have a smart meter from an Inclining Block tariff to a demand or time-of-use based tariff, and
- as a result of our review and assessment of customer assignment to ensure customers are assigned to the correct tariff class and tariff.

Ergon Energy initiated tariff re-assignments procedure for existing customers is further explained further in the sections below.

3.2.1 Tariff reassignment for SAC customers changing meter and/or with a smart meter

SAC customers with consumption below 100 MWh changing meter

End-of-life meter replacement for customers with a basic meter

Existing SAC Small residential and small business customers that have their basic accumulation meter replaced due to end of life reasons after 30 June 2020 may remain on the legacy Inclining Block tariffs for a period of 12 months from the date of the replacement.

At the end of this 12-month grace period, these customers will be reassigned to the applicable Residential or Small Business Transitional Demand tariff (RTDEM or BTDEM), unless their retailers have already voluntarily requested reassignment to a demand or time-of use network tariff prior to the end of the grace period.

Customer initiated meter upgrade from basic meter to smart meter

From 1 July 2020, existing SAC Small residential and small business customers that upgrade from a basic accumulation meter to a smart meter will be immediately assigned to the applicable Residential or Business Transitional Demand tariff (RTDEM or BTDEM).

SAC customers with consumption below 100 MWh with a smart meter

Existing residential and small business customers that already have their smart meter installed prior to 1 July 2020 may remain on the legacy Inclining Block tariffs until 1 July 2021.

On 1 July 2021 these customers will be reassigned to the applicable Residential or Small Business Transitional Demand tariff, unless their retailers have already voluntarily requested reassignment to a demand or time-of use network tariff prior to 1 July 2021.

Summary of SAC customer reassignment to tariffs after 1 July 2020

Table 4 below summarises tariff reassignment for SAC customers who change to a smart meter after 1 July 2020, and customers that already have their smart Type 4 meter installed prior to 1 July 2020.

Table 4: Reassignment of existing SAC customers after meter change

	Customer type	Existing tariff and tariff code	Tariff after reassignment	Reassignment date	Optional tariffs during 2020-21
SAC customers with consumption below 100 MWh changing to smart meter from 1 July 2020					
End-of-life meter failure	Residential	Residential Inclining Block - IBT	Residential Transitional Demand - RTDEM	12 months after meter replacement	Residential Inclining Block - RIB Residential Demand – RDEM Residential ToU Energy - RTOUE
	Small Business	Small Business Inclining Block - BIB	Small Business Transitional Demand - BTDEM	12 months after meter replacement	Small Business Inclining Block - BIB Small Business Demand – BDEM Small Business ToU Energy - BTOUE
Customer initiated action	Residential	Residential Inclining Block - IBT	Residential Transitional Demand - RTDEM	Immediately after meter change	Residential Inclining Block - RIB Residential Demand – RDEM Residential ToU Energy - BTOUE
	Small Business	Small Business Inclining Block - BIB	Small Business Transitional Demand - BTDEM	Immediately after meter change	Small Business Inclining Block - BIB Small Business Demand – BDEM Small Business ToU Energy - BTOUE
SAC customers with consumption below 100 MWh and smart meter installed prior 1 July 2020					
	Residential	Residential Inclining Block - IBT	Residential Transitional Demand - RTDEM	1 July 2021	Residential Inclining Block - RIB Residential Demand – RDEM Residential ToU Energy - RTOUE
	Small Business	Small Business Inclining Block - BIB	Small Business Transitional Demand - BTDEM	1 July 2021	Small Business Inclining Block - BIB Small Business Demand – BDEM Small Business ToU Energy - BTOUE

3.2.2 Periodic review and assessment

Ergon Energy will review the assignment of customers to tariffs to ensure customers are assigned to the correct tariff class and tariff. There are a number of circumstances where our review may identify that an existing customer is no longer eligible to remain assigned to their existing network tariff,

including when:

- CAC or ICC customers change in their voltage level of supply or a material change in connection assets to the extent that they are no longer able to remain on their existing tariff, or
- SAC customers have changed their usage of the electricity network to the extent that they are no longer eligible to remain assigned to their existing network tariff.

Electricity consumption levels for all eligible³ SAC customers are reviewed every twelve months in order to assess if their annualised consumption falls below/above the 100 MWh per year threshold. As a safeguard, a 15% tolerance limit is applied on an annualised consumptions basis to mitigate frequent tariff re-assignment.

For SAC Small customers with a smart meter that exceed the 100 MWh per year threshold we will initiate a network tariff change to reassign the customer to the Demand Small tariff (DST). SAC customers with basic metering who become SAC Large (i.e. consumption greater than 100 MWh per year) between 1 July 2020 and 30 June 2021 will remain on their legacy SAC Small network tariff (inclining block or WIFT) until they have upgraded to a smart meter. These customers will not be permitted to access any other SAC Small network tariffs. On 1 July 2021, these will be reassigned to new residential and business tariff specifically designed for these large, basic meter customers.

For SAC Large customers, where our review identifies that their annualised consumption is under the 100 MWh threshold, we will initiate a network tariff change to reassign the customer to the applicable SAC Small Transitional Demand tariff.

In accordance with our TSS, Ergon Energy will notify the retail customer prior to the proposed network tariff re-assignment occurring.

If a network tariff is discontinued or amended such that the tariff is no longer available to a customer, Ergon Energy may initiate a change to the customer's network tariff. This change will also be undertaken in accordance with procedures outlined in our TSS.

3.2.3 Retailer requested reassignment or reclassification

In accordance with our TSS, existing customers requesting a tariff re-assignment are allowed only one tariff change per 12-month period⁴. Such a tariff change is free of charge to customers.

For retailer initiated reclassification and network tariff code change process refer to our TSS, specifically Section 5.3.

3.3 Notice of proposed reassignment and objections review process

In accordance with our TSS, Ergon Energy will notify the retail customer or their retailer prior to the proposed network tariff re-assignment occurring to inform them about the proposed change, the reason for the change, how the customer can dispute the decision and the date the change will take effect.

³ Typically a NMI must have a minimum of six months of available consumption data in order to be reviewed

⁴ This condition will not apply to customers who have opted in to the newly introduced Small Business Primary Load Control Tariff, the Large Business Primary Load Control Tariff and the Large Business Secondary Load Control Tariff. Customers on these tariffs will be permitted to opt out of their load control tariffs within the 12-month period.

For new CAC and ICC customers: During the testing/commissioning stage of the connection process customers may be given the opportunity for a tariff reclassification within the 12-month period, in line with the requirements of the Customer Connection Agreement.

For further information about Ergon Energy's tariff reassignment process, including customer notification process and tariff assignment objection review refer to our TSS, Sections 5.4 and 5.5.

4. Description of network tariffs and application of charges

This chapter describes Ergon Energy's network tariffs to assist retailers, customers and other stakeholders to understand our Network Tariff Tables, particularly the tariff structures and the application of charging parameters.

4.1 Different types of network charges

Each network tariff comprises a combination of network charges (also referred to as charging parameters) that are applied to recover costs⁵. This section explains the different types of charging parameters used by Ergon Energy.

4.1.1 Network Use of System (NUOS) charges

Different types of NUOS charges and their application are described below.

Fixed charge

- A fixed \$/day charge applied to each energised connection point where energy or demand is recorded. In the case of the Wide Inclining Fixed tariff (WIFT), the fixed charge increases with each consumption block increment (refer to Section 4.3 and Appendix A for further details).
- In some situations, daily pro-rating will apply in the calculation and billing of fixed charges. The Queensland Market Participant Handbook provides further guidance on network billing arrangements.
- For small customers, fixed charges are designed to reflect the average capacity of the electricity network allocated to a typical customer on that network tariff. For large customers, fixed charges reflect the costs associated from the connection and management of the customer.

Volume charge

A volume charge may be a flat or variable charge for energy consumed at a connection point, calculated in \$/kWh:

- Flat volume charge - A flat or single volume charge, meaning the same price is charged for energy consumed regardless of when the energy is consumed. These charges are designed to recover the costs related to the volume (or amount) of electricity consumed by customers.
- Time of Use (ToU) volume charge - A variable volume charge, meaning the price charged for energy consumed changes at different times of the day. Prices are lower during Day (Off Peak) Hours and higher during Evening (Peak) Hours. Overnight (Shoulder) prices apply in-between the Evening and Day periods. These charges are designed to reduce demand on the

⁵ Network tariffs are applied to the electricity used at the connection point, as measured by the meter (or meters) at that connection point. Customers with multiple network connections will pay network charges for each connection point. This approach is consistent with the National Metering Identifier (NMI) Procedure issued by the Australian Energy Market Operator.

network during peak times by encouraging customers to switch non-essential electricity consumption to other periods.

Demand charge

- A monthly demand charge calculated as a \$/kVA/month or \$/kW/month for demand recorded at a connection point. These charges are applied to the maximum half hourly kW (or kVA for large customers) power reading that occurred at a connection point during either:
 - a single peak recorded anytime in the month, or
 - the maximum demand recorded within a peak demand window.

In some situations, daily pro-rating will apply in the calculation and billing of demand charges. The Queensland Market Participant Handbook provides further guidance on network billing arrangements.

- These charges are designed to reflect the costs associated with providing sufficient network capacity to a specific customer to cater for their maximum network demand. This means that customers who put more pressure on the network are charged more. As a result, these charges encourage customers to reduce their electricity costs by reducing their maximum demand.

Capacity Charge (only applicable to CAC and ICC customers)

- Capacity charge is a monthly charge calculated as a \$/kVA/month for the network capacity provided for a connection point.
- These charges assign an amount of shared network costs associated with providing network capacity that reflects the amount of capacity set aside for a specific customer and that can be used by that specific customer at any time.

4.1.2 Metering service charges

In addition to the NUOS charges, LV customers accessing Ergon Energy's network tariffs may be charged metering service charges⁶. Metering service charges are applied through a fixed \$/day charge. Metering charges are split into two components:

- a non-capital component that is applied to existing customers with legacy basic (Type 6) meters and continues to apply until a customer's meter is replaced with a smart Type 4 meter. This charge is designed to recover cost incurred in providing meter maintenance, reading, data services for basic meters.
- a capital component that is applied to existing customers connected to Ergon Energy's network prior to 1 July 2015, to recover the remaining capital cost related to legacy Type 6 meters⁷. This charge applies regardless of whether customers have upgraded to a smart meter or churned to an alternative meter provider.

The following types of capital and/or non-capital metering charges may be applied, subject to customer's metering:

⁶ Metering service charges classified as Alternative Control Services (ACS). For ACS, the cost of the service is not recovered through the NUOS charges. ACS are akin to a 'user-pays' system.

⁷ Note: Exemptions apply in the Power of Choice exempt areas (Mount Isa-Cloncurry and other Isolated supply networks) where Ergon Energy remain responsible for the installation and replacement of metering equipment.

- a charge for the primary metering service
- a supplementary charge for each secondary controlled load, and
- a supplementary charge for solar PV.

For further information about the application of metering service charges refer to our TSS Section 5.

4.2 Overview of charging parameters by tariff

The types of charging parameters that apply to Ergon Energy's 2020-21 network tariffs are shown in Table 5.

Table 5: Tariffs and their charging parameters

Network Tariff	Network tariff code ⁸	Fixed charge (\$/day)	Volume charge (\$/kWh)		Demand charge (\$/kw/month or S/kVA/month)		Capacity charge (\$/kVA)	Connect unit (\$/day/unit)	Metering services charge** (\$/day)
			Flat	Time-of-Use	Anytime	Peak window only			
Residential Inclining Block	RIB	√	√ Inclining						√
Residential Transitional Demand	RTDEM	√	√			√			√
Residential Demand	RDEM	√	√			√			√
Residential ToU Energy	RTOUE	√		√					√
Small Business Inclining Block	BIB	√	√ Inclining						√
Small Business Wide Inclining Fixed Tariff	BWIF	√ Inclining		√					√
Small Business Transitional Demand	BTDEM	√	√			√			√

⁸ It is intended that a network tariff code will apply to each meter data stream. In the case where a NMI has multiple meters (and data streams), metering data may be aggregated to calculate network charges. The Queensland Market Participant Handbook provides further guidance on how network tariffs are applied to aggregate data streams.

Network Tariff	Network tariff code ⁸	Fixed charge (\$/day)	Volume charge (\$/kWh)		Demand charge (\$/kw/month or S/kVA/month)		Capacity charge (\$/kVA)	Connect unit (\$/day/unit)	Metering services charge** (\$/day)
			Flat	Time-of-Use	Anytime	Peak window only			
Small Business Demand	BDEM	√	√			√			√
Small Business ToU Energy	BTOUE	√		√					√
Small Business Primary Load Control	BPLC	√	√						√
Volume Night Controlled	VN		√						√
Volume Controlled	VC		√						√
Demand Small	DST	√	√		√				√
Demand Medium	DMT	√	√		√				√
Demand Large	DLT	√	√		√				√
Seasonal ToU Demand	STOUD	√		√ Seasonal		√ Seasonal TOU			√
Large Business ToU Demand	LTOUD	√	√		√*	√			√
Large Business Primary Load Control	LPLC	√	√						√
Large Business Secondary Load Control	LSLC		√						√
Unmetered Supply	UM		√						√
33kV***	C33	√	√		√		√	√	

Network Tariff	Network tariff code ⁸	Fixed charge (\$/day)	Volume charge (\$/kWh)		Demand charge (\$/kw/month or \$/kVA/month)		Capacity charge (\$/kVA)	Connect unit (\$/day/ unit)	Metering services charge** (\$/day)
			Flat	Time- of-Use	Anytime	Peak window only			
66kV***	C66	√	√		√		√	√	
22/11kC Bus***	C22B	√	√		√		√	√	
22/11kV Line***	C22L	√	√		√		√	√	
Seasonal ToU Demand 11 or 22kV Bus***	C22BTOU T	√		√		√ Seasonal TOU	√	√	
Seasonal ToU Demand 11 or 22kV Line***	C22LTOU T	√		√		√ Seasonal TOU	√	√	
Seasonal ToU Demand 33 or 66kV***	C66TOU T	√		√		√ Seasonal TOU	√	√	
ICC tariff***	ICC	√	√		√		√		

* Excess demand charges may apply anytime outside the peak period. These charges are additional to the peak demand charges.

** Metering service charges may apply to some customers. Refer to Section 4.1.2 for further information about the application of metering service charge.

*** CAC and Standard ICC tariffs are not offered in Mount Isa region

Table 6: Tariffs and their charging timeframes

Network Tariffs	Charging timeframes	Weekdays ^a	Weekends
Residential ToU Energy	Evening (Peak) volume	4pm – 9pm	4pm – 9pm
	Overnight (Shoulder) volume	9pm – 9am	9pm – 9am
	Day (Off Peak) volume	9am – 4pm	9am – 4pm
Small Business ToU Energy	Evening (Peak) volume	4pm – 9pm	No peak charging
	Overnight (Shoulder) volume	9pm – 9am	4pm – 9am
	Day (Off Peak) volume	9am – 4pm	9am – 4pm
Residential Transitional Demand Residential Demand	Peak demand	4pm – 9pm	4pm – 9pm
Small Business Transitional Demand Small Business Demand	Peak demand	4pm-9pm	No peak charging
Large Business ToU Demand	Peak demand	4pm – 9pm	No peak charging
	Off-peak demand	9pm – 4pm	Anytime
Seasonal ToU Demand CAC Seasonal ToU Demand tariffs	Peak volume	All times during Summer ^b	All times during Summer ^b
	Peak demand	10am to 8pm during Summer ^b	10am to 8pm during Summer ^b
	Off-peak volume and demand	All times during non-Summer months	All times during non-Summer months

Notes:

- Weekdays include government gazetted full day public and bank holidays i.e. State, regional and local public holidays.
- 'Summer' is defined as the months of December, January and February.
- As agreed between the retailer and the customer one of the following periods: 7am to 7pm; 7.30am to 7.30pm; or 8am to 8pm.

4.3 Tariff specific information

4.3.1 Default Tariffs

Table 7: Default primary tariffs for SAC Small Residential customers

Tariff class: Standard Asset Customers (SAC)														
Customer Type:	Residential customer consuming up to 100 MWh per year													
Tariff:	Residential Inclining Block (Tariff code: RIB)													
Tariff description	This tariff has an inclining block structure, with the prices increasing with each step up in a customer's energy consumption (kWh) level above defined thresholds. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with any other primary residential tariff.													
Opt in / opt out arrangements	This tariff is the default tariff for residential customers with basic (Type 6) meters consuming up to 100 MWh per year. Arrangements for customers with a smart meter for 2020-21: - default tariff⁹ for residential customers with a smart meter where the customer's smart meter was installed before 1 July 2020, or during 2020-21 and upgraded from basic to smart metering for end of life replacement - Optional tariff for any other residential customers with a smart meter. Arrangements for customers with a smart meter during 2021-25: - default tariff for residential customers who upgraded from basic to smart metering for end of life replacement reasons in the previous 12 months. - not available to any other residential customers with a smart meter.													
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period Volume charge: inclining volume charge, \$/kWh, applies based on kWh energy usage in billing period. The volume charge is charged according to three blocks. The inclining blocks are triggered once a customer exceeds each nominated consumption threshold. For network billing and operational purposes, the IBT is denominated and applied on a daily basis. The annual equivalent kWh is provided for presentation purposes only. The calculation methodology for this tariff is set out in Appendix B. The following consumption blocks apply: <table><tr><th>Block</th><th>Daily kWh</th><th>Annual equivalent kWh</th></tr><tr><td>Block 1</td><td><2.74 kWh</td><td><1,000 kWh per year</td></tr><tr><td>Block 2</td><td>2.74 - 16.43 kWh</td><td>1,000 - 6,000 kWh per year</td></tr><tr><td>Block 3</td><td>>16.43 kWh</td><td>>6,000 kWh per year</td></tr></table>		Block	Daily kWh	Annual equivalent kWh	Block 1	<2.74 kWh	<1,000 kWh per year	Block 2	2.74 - 16.43 kWh	1,000 - 6,000 kWh per year	Block 3	>16.43 kWh	>6,000 kWh per year
Block	Daily kWh	Annual equivalent kWh												
Block 1	<2.74 kWh	<1,000 kWh per year												
Block 2	2.74 - 16.43 kWh	1,000 - 6,000 kWh per year												
Block 3	>16.43 kWh	>6,000 kWh per year												

⁹ Except for customers already assigned to a cost reflective tariff

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Residential customer consuming up to 100 MWh per year
Tariff:	Residential Transitional Demand (Tariff code: RTDEM)
Tariff description	The Transitional Demand tariffs for SAC Small customers are intended to be introductory demand tariffs for residential and small business customers and incorporate a lower demand price compared to the standard Demand tariffs, to allow customers to adjust to tariffs they may not be familiar with. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with Residential Inclining Block tariff.
Opt in / opt out arrangements	This tariff is the default for new residential customers, and for existing residential customers who initiate an upgrade to a smart meter, consuming up to 100 MWh per annum. Optional tariff for any other residential customer with a smart meter. From 1 July 2021, this tariff will also become the default tariff for: - residential customers who upgrade from basic to smart metering for end of life replacement reasons, with the default reassignment to this tariff occurring 12 months after smart meter installation, and - any other residential customers with smart metering who were assigned to the Residential Inclining Block as at 30 June 2021.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kW/month, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 4pm to 9pm weekdays and weekends

Table 8: Default primary tariffs for SAC Small Business customers

Tariff class: Standard Asset Customers (SAC)														
Customer Type:	Small business customer consuming up to 100 MWh per year													
Tariff:	Small Business Inclining Block (Tariff code: BIB)													
Tariff description	This tariff has an inclining block structure, with prices increasing with each step up in a customer’s energy consumption (kWh) level above defined thresholds. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with any other primary business tariff.													
Opt in / opt out arrangements	This tariff is the default tariff for small business customers with a basic (Type 6) meter consuming up and including 20 MWh per annum. Arrangements for customers with a smart meter for 2020-21: - default tariff¹⁰ for small business customers with a smart meter consuming up to and including 20 MWh per year where the customer’s smart meter was installed before 1 July 2020, or during 2020-21 and upgraded from basic to smart metering for end of life replacement. - Optional tariff for any other small business customers with a smart meter consuming less than 20 MWh per annum. Arrangements for customers with a smart meter during 2021-25: - default tariff for small business customers consuming up to and including 20 MWh per year who upgraded from basic to smart metering for end of life replacement reasons in the previous 12 months. - not available to any other small business customers with a smart meter.													
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period Volume charge: inclining volume charge, \$/kWh, applies based on kWh energy usage in billing period. The volume charge is charged according to three blocks. The inclining blocks are triggered once a customer exceeds each nominated consumption threshold. For network billing and operational purposes, the IBT is denominated and applied on a daily basis. The annual equivalent kWh is provided for presentation purposes only. The calculation methodology for this tariff is set out in Appendix B. The following consumption blocks apply: <table><tr><th>Block</th><th>Daily kWh</th><th>Annual equivalent kWh</th></tr><tr><td>Block 1</td><td><2.74 kWh</td><td><1,000 kWh per year</td></tr><tr><td>Block 2</td><td>2.74 - 54.76 kWh</td><td>1,000 - 20,000 kWh per year</td></tr><tr><td>Block 3</td><td>>54.76 kWh</td><td>>20,000 kWh per year</td></tr></table>		Block	Daily kWh	Annual equivalent kWh	Block 1	<2.74 kWh	<1,000 kWh per year	Block 2	2.74 - 54.76 kWh	1,000 - 20,000 kWh per year	Block 3	>54.76 kWh	>20,000 kWh per year
Block	Daily kWh	Annual equivalent kWh												
Block 1	<2.74 kWh	<1,000 kWh per year												
Block 2	2.74 - 54.76 kWh	1,000 - 20,000 kWh per year												
Block 3	>54.76 kWh	>20,000 kWh per year												
Tariff:	Small Business Wide Inclining Fixed Tariff (WIFT) (Tariff code: BWIF)													
Tariff description	This tariff is the default tariff for small business customers with a basic (Type 6) meter consuming more than 20 MWh and up to 100 MWh per annum. Secondary load control tariffs can be assessed with this primary tariff.													
Opt in / opt out arrangements	Closed to customers with a smart meter.													

¹⁰ Except for customers already assigned to a cost reflective tariff

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Small business customer consuming up to 100 MWh per year
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period This tariff has five inclining fixed charge blocks. To select the applicable fixed charge, a customer is assigned to one of the five blocks depending on their electricity usage i.e. different prices apply to each 20 MWh/year block. Block 1: Annual consumption up to 20 MWh/year Block 2: Annual consumption 20 MWh/year up to 40 MWh/year Block 3: Annual consumption 40 MWh/year up to 60 MWh/year Block 4: Annual consumption 60 MWh/year up to 80 MWh/year Block 5: Annual consumption equal to or exceeding 80 MWh/year The higher the customer's energy consumption, the higher the \$/day fixed charge. For further information refer to Appendix A. Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
Tariff:	Small Business Transitional Demand (Tariff code: BTDEM)
Tariff description	The Transitional Demand tariffs for SAC Small customers are intended to be introductory demand tariffs for residential and small business customers and incorporate a lower demand charge compared to the standard Demand tariffs to allow customers to adjust to tariffs they may not be familiar with. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with Small Business Inclining Block tariff.
Opt in / opt out arrangements	This tariff is the default for new small business customers and existing small business customers who initiate an upgrade to a smart meter, consuming up to 100 MWh per annum. Optional tariff for any other smart meter upgrade scenarios for small business customers. From 1 July 2021, this tariff will also become the default tariff for: - small business customers who upgrade from basic to smart metering for end of life replacement reasons, with the default reassignment to this tariff occurring 12 months after smart meter installation, and - any other small business customers with smart metering who were assigned to the Small Business Inclining Block as at 30 June 2021.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period Demand charge: A monthly charge calculated as \$/kW/month, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 4pm to 9pm weekdays (Note: Demand charges don't apply on weekends)

Table 9: Default tariff for SAC Large customers

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Large customers consuming 100 MWh or above per year
Tariff:	Large Business Time of Use Demand (Tariff code: LTOUD)
Tariff description	This tariff has time of use demand charges which apply during the peak demand window, and additional (excess) demand charges which may apply outside the peak window depending on the customers load characteristics. Customers must have appropriate metering to access this tariff as the demand charges are applied to the maximum half hourly kVA (Note: kW-based version of this tariff is not available).
Opt in / opt out arrangements	This tariff is the default tariff for new SAC large customers (consuming 100 MWh or above per year) with demand above 450 kVA. Optional tariff for all existing SAC large customers with a smart meter.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kVA/month, based on the maximum kVA demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 4pm to 9pm weekdays
	Excess demand charge: A monthly charge calculated as \$/kVA/month. It is measured as the single maximum demand outside the peak demand window minus the maximum demand during the peak demand window. Where the maximum monthly demand outside the peak demand window is less than the highest monthly maximum demand inside the peak window, the excess demand charge for that billing period is set to zero.
Tariff:	Demand Medium (Tariff code: DMT) Demand Small (Tariff code: DST)
Tariff description	The Demand Medium and Demand Small tariffs are anytime demand tariffs (i.e. these tariffs do not have a peak charging window for demand). The two tariffs have the same structure, however different rates and different thresholds for the application of the demand charges. The Demand Small and Medium tariffs are self-selecting with the customer determining the optimum tariff category based on their energy use and load characteristics.
Opt in / opt out arrangements	Default tariffs for SAC large customer with a smart meter consuming 100 MWh or above per year: Demand Small – default for new customers with demand greater than 30 kW (or 35 kVA) Demand Medium – default for new customer with demand greater than 120 kW (or 135 kVA) Optional tariffs for existing SAC large customers with a smart meter consuming 100 MWh or above per year. Note: Existing SAC Small Business and Residential customers with appropriate smart metering and consumption of 100 MWh or above per year, will be assigned by default to the Demand Small (DSL) tariff.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Large customers consuming 100 MWh or above per year
	Demand charge: A monthly charge calculated as \$/kVA/month (or kW)), applied to the kVA amount by which a customer's actual monthly maximum demand measured as a single peak over a 30-minute period during the month is greater than the demand threshold applicable to the customer's network tariff. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero and no demand charge is payable for that month. The threshold demands applicable to the SAC Large demand tariffs are: Demand Medium: 120 kW or 135 kVA Demand Small: 30 kW or 35 kVA Note: Where the customer has appropriate metering, demand charges will be kVA denominated. Where customer metering does not support kVA billing data being available, a kW denominated version of the tariff will continue to be available.

Table 10: Unmetered supply tariff

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Unmetered supplies for facilities
Tariff:	Unmetered (Tariff code: UM)
Tariff description	This tariff is available for small uniform loads that have no meter at the connection point, such as public lighting, traffic lights, security lights and other types of unmetered public amenities (e.g. illuminated signs, phone boxes and public barbecues). Ergon Energy only provides connection to network for these services.
Opt in / opt out arrangements	The unmetered supply network tariff applies to all loads approved to be unmetered by Ergon Energy¹¹. No other tariffs are available for unmetered supplies.
Charging parameter and application	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period

Table 11: Default CAC tariffs

Tariff class: Connection Asset Customers (CAC)	
Customer Type:	Customers with a network coupling point at 66 kV, 33 kV, 22 kV, 11 kV and installed capacity above 1,000 kVA
Tariffs:	66kV (Tariff code: C66) 33kV (Tariff code: C33) 22/11 kV Bus (Tariff code: C22B) 22/11 kV Line (Tariff code: C22L)

¹¹ The NER prescribes which metering installations do not require a meter (Type 7)

Tariff class: Connection Asset Customers (CAC)	
Customer Type:	Customers with a network coupling point at 66 kV, 33 kV, 22 kV, 11 kV and installed capacity above 1,000 kVA
Tariff description	These tariffs are designed for large commercial and industrial customers, typically with demand 1,000 kVA and above. Customers of CAC tariffs have site specific aspects which are required to calculate network charges (e.g. authorised demand, number of connection units).
Opt in / opt out arrangements	Default for new and existing CAC customers connected with the appropriate network coupling point
Charging parameter and application	Fixed charge: \$/day
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kVA/month, based on the maximum kVA demand measured as a single peak over a 30-minute period during the month.
	Capacity charge: Capacity charge is the greater of the authorised kVA demand or maximum kVA demand recorded in any 30-minute period during the billing month. Where CACs have an alternate supply (in addition to their primary supply), the authorised kVA demand will be set at zero for the alternate supply.
	Connection unit charge: Connection unit charges apply for customers who have connected to our network under legacy arrangements, prior to 1 July 2010. The DUOS connection unit calculation multiplies the connection unit charge (\$/day) by the customer's site-specific number of connection units. Refer to Appendix D for example.

Table 12: ICC tariff

Tariff class: Individually Calculated Customers (ICC)	
Customer Type:	Customers assigned to the ICC tariff class
Tariffs:	Standard ICC tariff and Non-Standard ICC tariff (Tariff code: ICC)
Tariff description	ICC tariffs are site specific and are calculated on an individual basis to reflect the specific site's load requirements. ICC tariffs are confidential – they are provided directly to the customers and/or the customer's retailer (they are not published on our website). Standard ICC tariff are not offered in the Mount Isa region. Non-Standard ICC tariff is only available to eligible existing CAC customers that satisfy the criteria for the ICC tariff in accordance with Appendix A of the TSS. The price level of the DUOS peak demand charging parameter of a non-standard ICC tariff is transitioned to Long run marginal cost over a reasonable period of transition to mitigate the impact of the introduction of cost reflective pricing.
Opt in / opt out arrangements	All customers classified as an ICC must be on a site-specific ICC tariff. No other tariff options are available.

Tariff class: Individually Calculated Customers (ICC)	
Customer Type:	Customers assigned to the ICC tariff class
Charging parameter and application	Fixed charge: \$/day - These charges vary for each customer depending on the customer's connection assets and funding arrangements. Connection assets are the assets required to connect an electrical installation to the shared network and are all the assets from the connection point back up to and including the network coupling point. Dedicated connection assets are generally for the sole use of a single connection and are typically not shared by multiple connections. In circumstances where the network coupling point, and/or identification of dedicated connection assets, is unclear or contested, Ergon Energy will consider other information, including but not limited to, the customer's metering point to make a determination about the network coupling point.
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kVA/month, based on the maximum kVA demand measured as a single peak over a 30-minute period during the month.
	Capacity charge: \$/kVA/month
	The capacity charge is the greater of the authorised demand (in kVA) or on the maximum kVA demand measured as a single peak over a 30-minute period during the month.

4.3.2 Optional tariffs

SAC optional tariffs

Table 13: SAC Small Residential customer optional primary tariffs

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Residential customer consuming up to 100 MWh per year
Tariff:	Residential Demand (Tariff code: RDEM)
Tariff description	This is a demand based tariff, designed to encourage residential customers to reduce their electricity costs by reducing their maximum demand during the peak times. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with Residential Flat.
Opt in / opt out arrangements	This tariff is optional for new and existing residential customers with a smart meter consuming up to 100 MWh per annum.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh/month, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kW, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 4pm to 9pm weekdays and weekends
Tariff:	Residential Time of Use (ToU) Energy (Tariff code: RTOUE)
Tariff description	This is a tariff with rates varying depending on the time of day. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with Residential Flat.
Opt in / opt out arrangements	This tariff is optional for new and existing residential customers with a smart meter consuming up to 100 MWh per annum.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A variable charge, calculated in \$/kWh, with different rates applying to the energy used at a connection point at different times of the day
	The following time periods apply to volume charges: Evening (peak): 4pm to 9pm on weekdays and weekends Night (shoulder): 9pm to 9am on weekdays and weekends Day (off-peak): 9am to 4pm on weekdays and weekends

Table 14: SAC Small Business customer optional primary tariffs

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Small business customer consuming up to 100 MWh per year
Tariff:	Small Business Demand (Tariff code: BDEM)
Tariff description	This is a demand based tariff, designed to encourage small business customers to reduce their electricity costs by reducing their maximum demand during the peak times. Secondary load control tariffs can be assessed with this primary tariff. This tariff cannot be used in conjunction with Small Business Flat.
Opt in / opt out arrangements	This tariff is optional for new and existing small business customers with a smart meter consuming up to 100 MWh per annum.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kW/month, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 4pm to 9pm weekdays (Note: Demand charges don't apply on weekends)
Tariff:	Small Business Time of Use (ToU) Energy (Tariff code: BTOUE)
Tariff description	This is a tariff with rates varying depending on the time of day, and with an inclining fixed charge structure. Secondary load control tariffs can be assessed with this primary tariff.
Opt in / opt out arrangements	This tariff is optional for new and existing residential customers with a smart meter consuming up to 100 MWh per annum.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period. This tariff has five inclining fixed charge blocks. To select the applicable fixed charge, a customer is assigned to one of the five blocks depending on their electricity use i.e. different rates apply to each 20MWh/year block). Block 1: Annual consumption up to 20 MWh/year Block 2: Annual consumption 20 MWh/year up to 40 MWh/year Block 3: Annual consumption 40 MWh/year up to 60 MWh/year Block 4: Annual consumption 60 MWh/year up to 80 MWh/year Block 5: Annual consumption equal to or exceeding 80 MWh/year The higher the customers annual energy consumption, the higher the \$/day fixed charge.
	Volume charge: A variable charge, calculated in \$/kWh, with different prices applying to the energy used at a connection point at different times of the day The following time periods apply to volume charges: Evening (peak): 4pm to 9pm on weekdays Night (shoulder): 9pm to 9am on weekdays; 4pm to 9am on weekends Day (off-peak): 9am to 4pm on weekdays and weekends
Tariff:	Small Business Primary Load Control Tariff (Tariff code: BPLC)
Tariff description	On this tariff electricity supply will be available for a minimum of 18 hours per day during time periods set at the absolute discretion of Ergon Energy.
Opt in / opt out arrangements	This tariff is optional for eligible small business customers with a basic or smart meter consuming up to 100 MWh. For terms and conditions of this tariff refer to Ergon Energy's 2020-21 Pricing Proposal.

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Small business customer consuming up to 100 MWh per year
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period

Table 15: SAC Large customer optional primary tariff

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	Large customers consuming 100 MWh or above per year
Tariff:	Demand Large (Tariff code: DLT)
Tariff description	The Demand Large tariff is an anytime demand tariffs (i.e. this tariff does not have a peak charging window for demand). The tariff has the same structure as the Demand Small and Demand Medium tariffs, however, the prices and the application of demand charges differ.
Opt in and opt out arrangements	Optional tariff for new and existing SAC large customers with a smart meter consuming 100 MWh or above per year.
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period
	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
	Demand charge: A monthly charge calculated as \$/kVA/month (or kW), applied to the kVA amount by which a customer's actual monthly maximum demand measured as a single peak over a 30-minute period during the month is greater than the demand threshold applicable to the customer's network tariff. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero and no demand charge is payable for that month. The threshold demands applicable to the SAC Large demand tariff is: Demand Large: 400 kW or 450 kVA Note: Where the customer has appropriate metering, demand charges will be applied to the maximum half hourly kVA. Where customer metering does not support kVA billing data being available, a kW denominated version of the tariff will continue to be available.

Secondary Tariffs for SAC customers

Secondary tariffs can generally only be accessed in conjunction with a primary tariff. For example, a residential customer, in addition to their primary tariff, may elect to have some appliances (e.g. hot water system) subject to a secondary 'controlled load' network tariff. Secondary tariffs are only available to SAC customers. Available secondary tariffs are described in the table below.

Table 16: Secondary tariffs

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	SAC Small Residential and Small business customer consuming up to 100 MWh per year
Tariff:	Volume Controlled (Tariff code: VC)

Tariff class: Standard Asset Customers (SAC)	
Customer Type:	SAC Small Residential and Small business customer consuming up to 100 MWh per year
Tariff description	Specified connected appliances ¹² are controlled by network equipment so supply will be permanently available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Ergon Energy. This tariff can be used in conjunction with any primary SAC small tariff, except Small Business Primary Load Controlled tariff.
Opt in / opt out arrangements	This tariff is available for eligible new and existing customers with basic or smart meters. For terms and conditions of this tariff refer to Ergon Energy's Pricing Proposal.
Charging parameter and application	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
Tariff:	Volume Night Controlled (Tariff code: VN)
Tariff description	Specified connected appliances are controlled by network equipment so supply will be permanently available for a minimum period of 8 hours per day during time periods set at the absolute discretion of Ergon Energy. This tariff can be used in conjunction with any primary SAC small tariff, except Small Business Primary Load Controlled tariff.
Opt in / opt out arrangements	This tariff is available for eligible new and existing customers with basic or smart meters. For terms and conditions of this tariff refer to Ergon Energy's Pricing Proposal.
Charging parameter and application	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period
Customer Type:	Large customers consuming 100 MWh or above per year
Tariff:	Large Business Secondary Load Control Tariff (Tariff code: LSLC)
Tariff description	Total connected load is controlled by network equipment so supply will be permanently available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Ergon Energy.
Opt in / opt out arrangements	This tariff is available for eligible new and existing customers with basic or smart meters consuming 100 MWh or above per year. For terms and conditions of this tariff refer to Ergon Energy's Pricing Proposal.
Charging parameter and application	Volume charge: A flat volume charge, \$/kWh, applies based on kWh energy usage in billing period

¹² Approval of equipment to connect to controlled load network tariffs is at the absolute discretion of Ergon Energy. Where Ergon Energy's load control equipment exists, this may not be disconnected without Ergon Energy's prior written consent.

CAC optional tariffs

Table 17: CAC optional tariffs

Tariff class: Connection Asset Customers (CAC)	
Tariff:	Seasonal Time of Use Demand 11 or 22 kV Bus (Tariff code: C22BTOUT) Seasonal Time of Use Demand 11 or 22 kV Line (Tariff code: C22LTOUT) Seasonal Time of Use Demand 33 or 66 kV (Tariff code: C66TOUT)
Tariff description	The structure of these tariffs includes both a seasonal and time-of-day dimension, with a peak demand charge, an off-peak volume charge and a capacity charge (off-peak demand). These tariffs also have a connection unit charge.
Opt in / opt out arrangements	New and existing CAC customers may opt in to these tariffs, subject to customer's network coupling point and connection assets.
Charging parameter and application	Fixed charge: \$/day applies
	Volume charge: A variable charge, calculated in \$/kWh, with different prices applying to the energy used at a connection point at different seasons (times of the year). The following time periods apply to volume charges: Peak: all times during Summer months of December, January and February Off-peak: all times during non-summer months
	Demand charge: A monthly charge calculated as \$/kVA/month, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 10am to 8:00pm during Summer (months of December, January and February) weekdays
	Capacity charge: The DUOS off-peak capacity charge calculation uses the maximum of authorised kVA demand or the monthly actual kVA maximum demand during the off-peak window, which is all times during non-summer months and all times during Summer months excluding demands occurring during the peak window of 10am to 8pm on Summer weekdays. The TUOS capacity charge doesn't have seasonal or time-of use dimensions and uses the greater of the authorised kVA demand or any time maximum kVA demand recorded in any 30-minute period during the billing month.
	Connection unit charge: Connection unit charges apply for customers who have connected to our network under legacy arrangements, prior to 1 July 2010. The DUOS connection unit calculation multiplies the connection unit charge (\$/day) by the customer's site-specific number of connection units. Refer to Appendix D for example.

4.3.3 Closed (grandfathered) tariffs

Table 18: Closed SAC tariffs

Tariff class: Standard Asset Customers (SAC)	
Tariff:	Seasonal Time of Use Demand (Tariff code: STOUD)
Tariff description	For this tariff all the charging parameters, with the exception of the fixed charge, include both seasonal and time of day dimensions.
Opt in / opt out arrangements	This tariff is limited to existing SAC Large customers who were assigned to this tariff as at 30 June 2020. Existing customers will be able to remain on this tariff and, should they choose to, will be able to request to be reassigned to the Large Business ToU Demand tariff, or applicable anytime demand tariff (Demand Large, Demand Medium and Demand Small).
Charging parameter and application	Fixed charge: \$/day applies to each energised connection point for each day in billing period Volume charge: A variable charge, calculated in \$/kWh, with different prices applying to the energy used at a connection point at different seasons (times of the year). The following time periods apply to volume charges: Peak: all times during summer months of December, January and February Off-peak: all times during non-summer months None: The off-peak volume charge is higher than the peak charge. The peak DUOS volume charge is set to \$0/kWh. The TUOS and Jurisdictional Schemes volume charge is same all year around. Demand charge: A monthly charge calculated as \$/kW/month, based on the maximum kW demand measured as a single peak over a 30-minute period during the peak demand charging window/timeframe. Peak demand window: 10:00am to 8:00pm on summer (December, January and February) weekdays Off-peak demand window: All times during non-summer months Thresholds above which demand charges apply: Peak: 20kW Off-peak: 40kW The demand charge will be applied to the kW amount by which a customer's actual monthly maximum demand is greater than the demand threshold applicable to the relevant peak /off-peak period. Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero

5. Authorised Demand for CAC and ICC customers

- Where there is no connection agreement in place defining an ICC or CAC's Authorised Demand (AD), Ergon Energy will determine an AD value to apply in network tariff calculations. Generally this will be based on the annual maximum demands in the previous full pricing period prior to the setting of prices. Under certain circumstances, a more recent demand may be substituted (e.g. where there has been a significant change in demand after the previous full pricing period).
- Where there is no connection agreement, or where the connection agreement defines a ICCs or CACs Authorised Demand in kW, Ergon Energy will convert the Authorised Demand to a kVA measure as part of our annual price setting process. This conversion uses a compliant Power factor. Details of the customer's Authorised Demand in kVA will be set out in the customer's site-specific tariff schedule.
- For CACs that have a primary and alternate supply (as deemed by Ergon Energy), the Authorised Demand will be set to zero on the alternate supply NMI for the purpose of calculating TUOS capacity charges. Ergon Energy will also waive the TUOS fixed charge on the alternate supply NMI.

6. Distribution Loss Factors

6.1 Background

Distribution Loss Factors (DLFs) are used to increase the customer's meter energy amount to account for electrical losses for electricity transmitted on a distribution network (between a distribution network connection point and a transmission network connection point). The DLF generally varies depending on the location and voltage of a customer's connection point.

The National Electricity Rules require Ergon Energy to calculate DLFs annually, for each network tariff. DLFs are approved by the AER and published by the Australian Energy Market Operator on their website.

6.2 DLF categories

Average DLFs are calculated for each significant supply level in the network, however the DLFs for major customers are calculated individually to determine the losses directly attributable to their loads. The table below outlines the DLF categories used by Ergon Energy and their applications.

Table 19: DLF categories and applications

Category	Description	Application
Sub-transmission Bus	Applicable to connection points that are High Voltage metered at a Sub-transmission Bus with a voltage greater than 30 kV.	NMIs that are supplied from a zone substation by dedicated, greater than 30 kV lines and metered at or immediately adjacent to the zone substation would be eligible for a Sub-transmission Bus DLF. Zone substations are defined as substations where the voltage level is stepped down from a voltage greater than 30 kV.
Sub-transmission Line	Applicable to connection points that are High Voltage metered at a Sub-transmission Line with a voltage greater than 30 kV.	The Sub-transmission Line DLF will apply to all NMIs that are: - connected to high voltage sub-transmission lines (greater than 30 kV) - metered at the same voltage as the line is energised. If customers believe that the NMI would qualify for a high voltage Bus DLF rather than a high voltage Line DLF they should submit details justifying their claim to Ergon Energy. As an example, NMIs supplied from a zone substation by dedicated 22/11 kV lines and are metered at or immediately adjacent to the zone substation would be eligible for a 22/11 kV Bus DLF.
22/11kV Bus	Applicable to connection points that are High Voltage metered at a 22/11 kV Bus with a voltage less than 30 kV and greater than 1,000 volts.	As per "Sub-transmission Bus" application above.
22/11kV Line	Applicable to connection points that are High Voltage metered at a 22/11 kV Line with a voltage less than 30 kV and greater than 1,000 volts.	The 22/11 kV Line DLF will apply to all NMIs that are: - connected to high voltage distribution lines - metered at the same voltage as the line is energised.
Low Voltage (LV) Bus	Applicable to connection points that are Low Voltage metered at a Low	NMIs which are supplied from a distribution substation on their site (owned or leased) and

Category	Description	Application
	Voltage Bus with a voltage less than 1,000 volts.	metered at the low voltage level are eligible for a LV Bus DLF.
Low Voltage (LV) Line	Applicable to connection points that are Low Voltage metered at a Low Voltage Line with a voltage less than 1,000 volts.	All NMIs not covered by any of the above categories. If customers believe that the NMI would qualify for a LV Bus DLF rather than a LV Line DLF they should submit details justifying their claim to Ergon Energy.

For our SAC tariffs, the DLFs approved to apply are published on our website in our Network Tariff Tables. These DLFs are based on the most common network supply configuration. For our CAC and ICC tariffs, the DLFs are site specific and confidential. These DLFs are provided directly to the customers and/or the customer's retailer on request.

Ergon Energy's methodology for calculating DLFs is available on our website:

https://www.ergon.com.au/_data/assets/pdf_file/0012/596928/Distribution-Loss-Factor-Methodology.pdf

6.3 Application of DLFs

Ergon Energy applies the DLF to customers metered energy usage for the calculation of TUOS volume charges. That is, the customers actual consumption is 'uplifted' by the DLF value, then the resulting consumption value is multiplied by the published TUOS volume rate (\$/kWh) for the relevant tariff¹³. The adjustment allows application of the DLF for charging TUOS volume at the NMI level, rather than at a standardised tariff level.

The example below illustrates how DLFs are applied by Ergon Energy for billing purposes.

Example:

Tariff: Residential Transitional Demand (RTDEM) – Pricing zone East, Transmission zone 1

Monthly Metered Volume: 5,500 kWh

Charges:

Fixed:

No adjustment required for DLFs

ie. NUOS \$/day rate x number of days in billing period = fixed charge

Volume:

DUOS - Volume charge: 5,500 kWh x \$0.04233 = \$232.815

Jurisdictional Schemes – Volumes charge: 5,500 kWh x \$0.00822 = \$45.21

TUOS - Volume charge 5,500 kWh x DLF 1.093 = 6,011.5 uplifted kWh x \$0.00761 = \$45.75

Total - Volume charge = \$232.815 + \$45.21 + \$45.74 = \$314.77

Demand:

No adjustment required for DLFs

¹³ That is, the NUOS volume prices published in the Network Tariff Tables are not used by Ergon Energy for billing purposes - the TUOS volume component is first adjusted by the applicable DLF

ie. NUOS peak demand charge rate (\$/kW/month) x peak demand for the month (maximum kW demand measured as a single peak over a 30-minute period, in the peak demand charging window) = demand charge

7. Avoided TUOS payments to embedded generators

7.1 Background

In accordance with the National Electricity Rules (NER), Ergon Energy is required to pay Avoided Transmission Use of System (Avoided TUOS) to eligible Embedded Generators (EG) in Ergon Energy's distribution network. Avoided TUOS payments recognise that energy supplied to the electricity distribution network by the embedded generator would have otherwise been supplied from the transmission network.

Generally, to be eligible for Avoided TUOS payments the EGs must have:

- sought access to Ergon Energy's distribution network under Chapter 5 of the NER,
- a generator Connection Agreement with Ergon Energy and
- registered or intend to register with AEMO as a *Generator Rules Participant*.¹⁴

If an exemption applies, or there is no intention for the EG to register as a Generator Rules Participant, we will not make Avoided TUOS payments.

In specific circumstances, Avoided TUOS payments may be allowed to be received by another entity other than the EG (for example where an intermediary is appointed and registered as a Generator under the NER).

7.2 Methodology for calculating avoided TUOS

In accordance with the NER, to calculate the avoided TUOS payments for eligible EGs, we:

- (a) Determine the charges for the locational component of prescribed DPPC services that would have been payable by Ergon Energy had the EG not injected any energy at its connection point during that financial year.
- (b) Determine the amount by which the charges calculated in (a) exceeds the amount for the locational component of prescribed DPPC services actually payable by Ergon Energy.
- (c) Credit the value from (b) to the EG account.

Reverse flow and net load

Where electricity produced by the generator flows back into the transmission network at the transmission connection point (TCP), this is known as excess export, or reverse flow. Where there is reverse flow at the TCP level, that generation does not reduce our net load downstream of that TCP. Accordingly, we remove the reverse flowing electricity from the calculations of Avoided TUOS. This means, our calculation of Avoided TUOS for a particular EG will be based on the difference between:

- 1) The actual net load at the TCP (and the relevant locational component of prescribed TUOS charges) and
- 2) The net load at the TCP if the EG was not there (and the relevant locational component of prescribed TUOS charges).

In the event that multiple EGs are connected to the same TCP, and there is reverse flow through the TCP, Ergon Energy will apportion the reverse flow attributable to each EG in line with the proportion of each EG's generation into the distribution network. For example, if Generator A exports 100 MWh in a month and Generator B exports 200 MWh in a month, and there is 30 MWh of excess

¹⁴ Some embedded generating units are required to register as a Generator Rules Participant under the NER.

export/reverse flow into the transmission network in that month, we will attribute 10 MWh to Generator A ($100/300 \times 30 = 10$) and 20 MWh to Generator B ($200/300 \times 30 = 20$).

Avoided TUOS calculation

We use the below methodology to calculate Avoided TUOS:

- 1) determine the amount of energy sent out by the EG in the relevant financial year (kWh)
- 2) convert this to an equivalent amount of energy at the TCP, by adjusting the export energy by the DLF of the EG
- 3) determine the net generator output (i.e. the generator output that is utilised by the local distribution network, by subtracting the actual metered energy that flows back into the transmission network at the TCP). Where multiple generators are operating in the same local area, the reverse flow is apportioned to each EG using the principles outlined above.
- 4) add the net generation output to the TCP actual metered data for the financial year
- 5) determine the TUOS that would have been charged if the generator was not connected, by recalculating the customer TUOS usage charges (demand and energy)
- 6) subtract the actual TUOS payment from the amount calculated in step 5
- 7) arrange payment of the resultant value from step 6 to the EG (or intermediary).

7.3 Payment of Avoided TUOS

Avoided TUOS payments to EGs following the end of the relevant financial year will be made as agreed between Ergon Energy and the particular EG (or intermediary) and will generally be remitted in the form of a lump sum payment after 30 June 2021.

7.4 Recovery of Avoided TUOS

In accordance with the National Energy Rules, we are able to recover costs associated with Avoided TUOS through TUOS charges in our network tariffs. Where we are to pay an Avoided TUOS payment to an EG, the payment amount is recovered as part of the TUOS volume charges passed through to customers at the same connection point as the EG.

Appendix A: WIFT Fixed charge calculation methodology

The Small Business WIFT (tariff code: BWIF) is structured with five inclining blocks, each with a different \$/day fixed rate applicable and a flat volume rate (\$/kWh).

The WIFT tariff fixed charge blocks are:

	Annual consumption	Equivalent daily consumption kWh
Block 1	up to 20 MWh/year	Up to 54.79
Block 2	20 MWh/year up to 40 MWh/year	54.79 to 109.58
Block 3	40 MWh/year up to 60 MWh/year	109.58 to 164.38
Block 4	60 MWh/year up to 80 MWh/year	164.38 to 219.17
Block 5	equal to or exceeding 80 MWh/year	219.18

The WIFT fixed charge calculation methodology is as follows:

- 1) Calculate the total energy consumption (kWh) for the billing period
- 2) Calculate equivalent daily kWh value for the billing period = kWh consumption /number of days in billing period
- 3) Identify which 'Block' the customers daily kWh value fits in and select the corresponding fixed charge rate from the Network Tariff Tables

Multiply the value calculated in Step 2 by the number of days in the billing period = this is the total fixed charge for the billing period

Example:

A small business premises meter is read quarterly. On this occasion, the customers consumption is for the 5,000 kWh for that quarter's read.

Equivalent daily consumption = consumption divided by the number of days in the read =
 $5,000/90 = 55.55$ kWh per day

As 55.55 kWh is above Block 1 max threshold of 54.79 kWh but below Block 2 max threshold of 109.58 kWh, Block 2 should be selected

The inclining NUOS fixed charge = 90 days x 0.970 = \$87.3

The same methodology is applied for the calculation of the fixed charge component of the Small Business Time of Use Energy tariff (Tariff code: BTOUE).

(Note: The volume component of Small Business Time of Use Energy tariff is applied differently to the WIFT, as the ToU tariff has a variable volume charge)

Appendix B: IBT charge calculation methodology

The IBT is structured with a fixed rate (\$/day) and three volume consumption blocks, each with a different rate (\$/kWh).

Block sizes and prices are different between residential and business customers IBT.

The IBT may be described in the context of an annual basis, however it is denominated and applied on a daily basis for billing purposes. For example, the IBT Residential consumption blocks are:

Block	Daily kWh	Annual equivalent kWh
Block 1	<2.74 kWh	<1,000 kWh p.a.
Block 2	2.74 kWh – 16.43 kWh	1,000 kWh p.a. – 6,000 kWh p.a.
Block 3	>16.43 kWh	>6,000 kWh p.a.

Daily denomination ensures IBT billing is equitably applied for any meter reading period (including NMIs where a customer move-out/move-in occurs), based on an accumulated total of consumption divided by the number of days in the reading period.

The IBT network tariff calculation methodology is as follows:

- a meter read is taken, with total consumption for the number of days within that meter reading period
- for IBT network billing purposes, the energy (kWh) assigned to each block is prorated back to a daily equivalent for that meter reading period
- the bill is then calculated with the component parts being a daily fixed charge, Volume Charge Block 1, Volume Charge Block 2 and Volume Charge Block 3
- daily calculations are then converted back to the total network charge by multiplication by the number of days in the meter reading period
- in relation to TUOS – the TUOS volume component will effectively remain as a flat rate for all IBT tariffs (i.e. same TUOS rate applied in each tariff block or meter reading period) and the DLF is applied to the metered consumption for TUOS.

The IBT bill is calculated as follows:

Component	Calculation
Fixed Charge	Number of days in the period <i>multiplied by</i> fixed charge rate.
Volume Charge Block 1	If equivalent daily consumption is less than the Block 1 daily allowance then: equivalent daily consumption <i>multiplied by</i> Block 1 rate multiplied by days in meter reading period no further calculations required. Or, if equivalent daily consumption exceeds the Block 1 daily allowance then: Block 1 volume charge calculation applied proceed to Volume Charge Block 2 calculation.
Volume Charge Block 2	For consumption within the Block 2 allowance then: equivalent daily consumption <i>less</i> the Block 1 daily allowance <i>multiplied by</i> Block 2 rate (up to the max block 2 threshold) multiplied by days in meter reading period.
Volume Charge Block 3	For any consumption above the aggregate of Block 1 and Block 2 daily allowance then: remaining consumption above the Block 1 and Block 2 daily allowance <i>multiplied by</i> the Block 3 rate multiplied by days in meter reading period.

Example:

A residential customer is typically read quarterly. On this occasion, the customer is read at 90 days with the start read being 123,400 kWh and the end read as 125,200 kWh. This equates to a consumption of 1,800 kWh for that quarter's read.

Currently, the prices for each DUOS component (excluding GST) of the IBT Residential East tariff are as follows:

Fixed charge	Block 1 Rate	Block 2 Rate	Block 3 Rate
\$1.250	\$0.02194	\$0.05294	\$0.09069

Calculation

Equivalent Daily Consumption = consumption divided by the number of days in the read = $1,800/90 = 20.00$.

Fixed Charge = $90 \times \$1.250 = \112.50

Volume Charge Block 1 = $2.74 \times \$0.02194 \times 90 = \5.41

Volume Charge Block 2 = $(16.43 - 2.74) \times \$0.05294 \times 90 = \65.23

Volume Charge Block 3 = $(20 - 2.74 - 13.69) \times \$0.09069 \times 90 = \$29.17$

Quarterly DUOS = \$212.31

The second quarter for this customer shows a total consumption of 200 kWh over 88 days for the period.

Calculation

Equivalent Daily Consumption = consumption divided by the number of days in the read = $200/88 = 2.27$

Fixed Charge = $88 \times \$1.250 = \110.00

Volume Charge Block 1 = $2.27 \times \$0.02194 \times 88 = \4.38

Volume Charge Block 2 = $0 \times \$0.05294 \times 88 = \0

Volume Charge Block 3 = $0 \times \$0.09069 \times 88 = \0

Quarterly DUOS = \$114.38

Example:

A residential premise (e.g. a summer holiday beach) is typically read quarterly. On this occasion, the customer has the following consumption pattern.

	Meter reading period (days)	Consumption kWh
Quarter 1	90	1000
Quarter 2	88	0
Quarter 3	93	0
Quarter 4	95	0

The meter reading at the start of quarter 1 is 240,000 kWh and the reading at the end of the fourth quarter is 241,000 kWh.

Calculation – Quarter 1

Equivalent Daily Consumption = consumption divided by the number of days in the read = $1000/90 = 11.11$.
 Fixed Charge = $90 \times \$1.250 = \112.50
 Volume Charge Block 1 = $2.74 \times \$0.02194 \times 90 = \5.41
 Volume Charge Block 2 = $(11.11 - 2.74) \times \$0.05294 \times 90 = \39.88
 Volume Charge Block 3 = $(0) \times \$0.09069 \times 90 = \0
 Quarterly DUOS = \$157.79

Calculation – Quarter 2

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/88 = 00.00$.
 Fixed Charge = $88 \times \$1.250 = \110.000
 Volume Charge Block 1 = $(0) \times \$0.02194 \times 88 = \0
 Volume Block 2 = $(0) \times \$0.05294 \times 88 = \0
 Volume Block 3 = $(0) \times \$0.09069 \times 88 = \0
 Quarterly DUOS = \$110.00

Calculation – Quarter 3

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/93 = 00.00$.
 Fixed Charge = $93 \times \$1.250 = \116.25
 Volume Charge Block 1 = $(0) \times \$0.02194 \times 93 = \0
 Volume Charge Block 2 = $(0) \times \$0.05294 \times 93 = \0
 Volume Charge Block 3 = $(0) \times \$0.09069 \times 93 = \0
 Quarterly DUOS = \$116.25

Calculation – Quarter 4

Equivalent Daily Consumption = consumption divided by the number of days in the read = $0/95 = 00.00$.
 Fixed Charge = $95 \times \$1.250 = \118.75
 Volume Charge Block 1 = $(0) \times \$0.02194 \times 95 = \0
 Volume Charge Block 2 = $(0) \times \$0.05294 \times 95 = \0
 Volume Charge Block 3 = $(0) \times \$0.09069 \times 95 = \0
 Quarterly DUOS = \$118.75

Total Annual DUOS (excluding GST) = \$502.79

Appendix C: Seasonal TOU Demand tariff calculation examples

Note: All amounts in the worked examples below relate to DUOS only

CAC STOU Example:

Network Tariff Code	DUOS Charges (GST exclusive)				
	Connection Unit Charge	Fixed Charge	Capacity charge Off-peak	Actual Demand Charge Peak	Volume charge Off- peak
	\$/day per connection unit	\$/day	\$/kVA of AD/ month	\$/kVA/month	\$/kWh
EC66TOUT1	\$9.209	\$0.000	\$5.535	\$11.000	\$0.00370

Example:

Where the customer's:

AD = 4,000 kVA

Connection Units = 0

Actual Demand Peak for the month = 3,600 kVA

Actual Demand Off-Peak for the month = 3,900 kVA

Energy for the month = 1,600,000 kWh

Calculation – Month of January (Summer)

Connection Unit Charge = $\$9.209 \times 31 \text{ days} \times 0 \text{ connection units} = \0.00

Fixed Daily Charge = \$0.00

Capacity Charge Off-peak = $\$5.535 \times 4,000 \text{ kVA} = \$22,140.00$

Actual Demand Charge Peak = $\$11.000 \times 3,600 \text{ kVA} = \$39,600.00$

Volume Charge Off-peak = Does not apply during a summer month

Total monthly DUOS = \$61,740.00

Calculation – Month of September (Non-summer)

Connection Unit Charge = $\$9.209 \times 30 \text{ days} \times 0 \text{ connection units} = \0.00

Fixed Daily Charge = \$0.00

Capacity Charge Off-peak = $\$5.535 \times 4,000 \text{ kVA} = \$22,140.00$

Actual Demand Charge Peak = Does not apply during a non-summer month

Volume Charge Off-peak = $\$0.00370 \times 1,600,000 \text{ kWh} = \$5,920.00$

Total monthly DUOS = \$28,060.00

SAC Large STOU Example:

Network Tariff Code	Threshold above which demand charge applies		DUOS charges (GST exclusive)				
			Fixed charge	Actual Demand charge		Volume charge Peak	Volume charge Off-peak
	Peak kW	Off-peak kW		Peak	Off-peak		
			\$/day	\$/kW/ month	\$/kW/ month	\$/kWh	\$/kWh
ESTOUDCT1	20	40	\$27.000	\$59.087	\$9.499	\$0.00000	\$0.02227

Example 1:

Where the customer's:

Peak demand for the month = 50 kW

Peak energy for the month = 20,000 kWh

Calculation – Month of February (Summer)

Fixed Charge = \$27.000 x 28 days = \$756.00

Actual Demand Charge Peak = \$59.087 x (50 kW – 20 kW) = \$1,772.61

Actual Demand Charge Off-peak = Does not apply during a summer month

Volume Charge Off-peak = Does not apply during a summer month

Total monthly DUOS = \$2,528.61

Example 2:

Where the customer's:

Off-peak demand for the month = 40 kW

Off-peak energy for the month = 25,000 kWh

Calculation – Month of July (Non-summer)

Fixed Charge = \$27.000 x 31 days = \$837.00

Actual Demand Charge Peak = Does not apply during a non-summer month

Actual Demand Charge Off-peak = \$9.499 x (40 kW – 40 kW) = \$0

Volume Charge Off-peak = \$0.02227 x 25,000 kWh = \$556.75

Total monthly DUOS = \$1,393.75

Appendix D: CAC connection unit charge examples

Note: Amounts included in the worked examples below relate to DUOS only

Network Tariff Code	Default DLF	DUOS charges (GST exclusive)				
		Connection Unit Charge	Fixed Charge	Capacity Charge	Actual Demand Charge	Volume Charge
	Value	\$/day / connection unit	\$/day	\$/kVA of AD per month	\$/kVA per month	\$/kWh
EC66T1	Site specific	\$9.209	\$121.200	\$3.283	\$2.388	\$0.00421

Example 1:

Where the customer's:

AD = 3,500 kVA

Connection Units = 11

Actual Demand for the month = 3,000 kVA

Energy for the month = 1,400,000 kWh

Calculation

Connection Unit Charge = $\$9.209 \times 30 \text{ days} \times 11 \text{ connection units} = \$3,038.97$

Fixed Charge = $\$121.200 \times 30 = \$3,636.00$

Capacity Charge = $\$3.283 \times 3,500 \text{ kVA} = \$11,490.50$

Actual Demand Charge = $\$2.388 \times 3,000 \text{ kVA} = \$7,164.00$

Volume Charge = $\$0.00421 \times 1,400,000 \text{ kWh} = \$5,894.00$

Total monthly DUOS = $\$31,223.47$

Example 2:

Where the customer's:

AD = 4,000 kVA

Connection Units = 0

Actual Demand for the month = 3,900 kVA

Energy for the month = 1,900,000 kWh

Calculation

Connection Unit Charge = $\$9.209 \times 30 \text{ days} \times 0 \text{ connection units} = \0

Fixed Charge = $\$121.200 \times 30 = \$3,636.00$

Capacity Charge = $\$3.283 \times 4,000 \text{ kVA} = \$13,132.00$

Actual Demand Charge = $\$2.388 \times 3,900 \text{ kVA} = \$9,313.20$

Volume Charge = $\$0.00421 \times 1,900,000 \text{ kWh} = \$7,999.00$

Total monthly DUOS = $\$34,080.20$

Appendix E: Ergon Energy's kVA calculation methodology

The four metering data quadrants recorded by interval meters are referred to as:

- kW_{Load} the real component of load power consumed by the customer over a given time period (Export kWh - Ei)
- kW_{Gen} the real component of power generated by the customer over a given time period (Import kWh - Bi)
- $kVAr_{Lag}$ (kilovolt-amperes-reactive-hours) is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is lagging the voltage waveform (Export kVarh - Qi)
- $kVAr_{Lead}$ is a measure of the reactive power which exists when the current and voltage are out of phase, where the current waveform is leading the voltage waveform (Import kVarh - Ki).

The kVA algorithm to be used by Ergon Energy is detailed below.

This "Standard" algorithm specifically excludes:

- incidental generation back into the Ergon Energy network
- any Leading VArS generated by customer equipment
- any contribution to lagging VArS by the generator

through modification of interval data as described below,

$$kVA = \sqrt{(kW_{Load})^2 + (kVAr_{Lag})^2} \text{ for each 30 minute interval at a connection point.}$$

For the purposes of network billing, the monthly maximum demand is the maximum kVA of the half hourly values for each interval during the relevant month, calculated as above.

Where Ergon Energy has agreed to the summation of the customer's demand at more than one metering installation for the purposes of determining a diversified demand charge, the billable demand is the vector summation of kW_{Load} and $kVAr_{Lag}$ at the individual metering installations.

That is, the chargeable demand is calculated for each interval as:

$$kVA = \sqrt{(kW_{Load}^{Meter\ 1} + kW_{Load}^{Meter\ 2} + \dots)^2 + (kVAr_{Lag}^{Meter\ 1} + kVAr_{Lag}^{Meter\ 2} + \dots)^2}$$

Modification of interval data explained

Within a metering interval, there may be both kW_{Load} and kW_{Gen} , where the real demand at a site swings through zero from load to generation or vice-versa. There may also be both $kVAr_{Lag}$ and $kVAr_{Lead}$, where the reactive power swings from lagging to leading power factor, or vice-versa. Depending upon the excitation level of the embedded generator, it will contribute $kVAr_{Lag}$ or $kVAr_{Lead}$.

With kVA charging for loads, if the load demand were to be directly calculated as

$$kVA = \sqrt{(kW_{Load})^2 + (kVAr_{Lag})^2} \text{ for each 30 minute interval, the } kVAr_{Lag} \text{ component may contain}$$

a contribution from the generator. This has the potential to increase the total kVA and kVAr and may create the monthly maximum load demand and a kVAr level that exceeds the permissible quantity. It is not the intent of load side kVA charging for demand to include this generator impact. Therefore, the generator's contribution to kVA charges for the load needs to be negated.

For the purposes of Ergon Energy network billing for loads, where an ICC or CAC is also an Embedded Generator, the interval data is modified so that in any 30 minute interval where $B_i \neq 0$, Q_i is made equal to 0.

With the half hourly interval values of kWLoad and kVArLag modified to remove generator contribution as described the monthly maximum load kVA charge will be based on the vector sum of E_i and Q_i interval data. A load that swings from export to import within an interval would receive a charge based only on the energy exported from the grid to the customer for the interval and would not include any kVArLag contribution by the generator. Where a peak period has been nominated by Ergon Energy the charge would be based on the load based maximum demand occurring during this period.

Appendix F: TUOS regions and transmission connection points

Ergon Energy's TUOS tariff structures are based on geographical transmission regions (e.g. T1, T2, T3 or T4). The transmission regions are based on groupings of Transmission Connection Points.

A list of all the transmission connection points, their assigned transmission node identifiers and their geographical TUOS region is provided in the table below.

TUOS regional indicators for CAC and SAC network tariffs		
Transmission Node Identifier	TUOS Region	Transmission Connection Point
Mt Isa	T4	Mount Isa
QALC	T2	Alligator Creek
QALH	T2	Louisa Creek
QASF	T2	Alan Sherriff
QBIL	T1	Biloela
QBLK	T1	Bulli Creek
QBNN	T2	Bowen North
QBWH	T1	Blackwater 132kV (Rolleston)
QBWL	T1	Blackwater 66 kV
QCAR	T1	Gladstone North (Calliope River)
QCBL	T1	Columboola
QCDW	T3	Cardwell
QCHA	T1	Chinchilla
QCLR	T2	Clare South
QCNS	T3	Cairns 132kV
QCOL	T2	Collinsville
QCRN	T3	Cairns 22 kV
QDGL	T2	Dan Gleeson
QDYS	T1	Dysart
QEGN	T1	Egans Hill
QELA	T3	El Arish
QEMS	T2	Kemmis
QEMT	T3	Edmonton
QGAR	T2	Garbutt
QGNG	T1	Gin Gin
QGST	T1	Gladstone South
QINF	T3	Innisfail
QING	T3	Ingham
QKAM	T3	Kamerunga
QKCK	T2	King Creek
QLCM	T1	Lilyvale 132kV
QLIL	T1	Lilyvale 66 kV
QMKA	T2	Mackay
QMRA	T1	Moura
QMRG	T1	Middle Ridge
QMRH	T1	Broadlea
QMRL	T1	Moranbah 11 kV
QMRN	T1	Moranbah 66 kV

TUOS regional indicators for CAC and SAC network tariffs		
QNEB	T2	Nebo
QNLD	T2	Newlands
QOKT	T1	Oakey
QPAL	T1	Pandoin
QPIV	T2	Pioneer Valley
QPRO	T2	Proserpine
QROC	T1	Rockhampton
QROS	T2	Ross
QSYC	T2	Stony Creek
QTBC	T1	Teebar Creek
QTKM	T1	Tangkam
QTLL	T3	Tully
QTRL	T1	Tarong 66 kV
QTUH	T3	Turkinje 132kV
QTUL	T3	Turkinje 66 kV
QTVE	T2	Townsville East
QTVS	T2	Townsville South
QWLN	T1	Woolooga
QWRE	T3	Woree
QWSH	T1	Wandoan South
QYAE	T1	Boat Creek

Appendix G: Glossary

Table 20: Definitions of terminology used throughout this document

Term	Abbreviation / Acronym	Definition
Australian Energy Regulator	AER	The economic regulator of the NEM established under section 44AE of the <i>Competition and Consumer Act 2010</i> (Commonwealth).
Alternative Control Service		Customer specific or customer requested services. These services may also have potential for provision on a competitive basis rather than by the local DNSP.
After hours		The maximum demand permitted to be imported or exported to the network by a network user, based on the nature of their connection.
Authorised demand		The maximum demand permitted to be imported or exported to the network by a network user, based on the nature of their connection.
Basic meter		Basic accumulation meters are defined as a meter that is only capable to recording the customers' energy consumption during the billing period.
Business hours	BH	8 am to 5 pm, Monday to Friday.
Capacity charge		A type of charge (charging parameter) included in network tariff structures. The capacity charge seeks to reflect the costs associated with providing network capacity required by a customer on a long term basis. It is levied on the basis of either contracted demand or forecasted capacity using prior year information.
Charging parameter		The charges comprising a tariff. Parameters include demand, capacity, fixed and volume (flat or time-of-use) charges.
Common service		A service that ensures the integrity of a distribution system, benefits all distribution customers and cannot reasonably be allocated on a locational basis.
Connection asset (Contributed or non-contributed)		Related to building connection assets at a customer's premises as well as the connection of these assets to the distribution network. Connection assets can be contributed (customer funded, then gifted to Ergon Energy) or non-contributed (Ergon Energy funded).
Connection point		The agreed point of supply established between a Network Service Provider and another Registered Participant, Non-Registered Customer or franchise customer. The meter is installed as close as possible to this location.
Customer		Refer to chapter 10 of the NER.
Demand		The amount of electricity energy being consumed at a given time measured in either kilowatts (kW) or kilovolt amperes (kVA). The ratio between the two is the power factor.
Demand charge		A type of charge (charging parameter) included in network tariff structures. This charge accounts for the actual demand a customer places on the electricity network. Different parameters apply to this charged depending on the different tariffs.
Demand tariff		The tariff has been structured to include a demand component, so the customer's actual demand is reflected in the price they pay for their electricity.
Designated Pricing Proposal Charge	DPPC	Refers to the charges incurred for use of the transmission network; previously referred to as Transmission Use of System (TUOS).
Distribution Use of System	DUOS	This refers to the network charges which recover the costs of providing Standard Control Services.
Embedded Generator	EG	In line with the ENA classification, EGs are generally those generators with an installed capacity as follows: Medium: 1-5 MVA (LV or HV) or < 1 MVA (HV) Large: > 5 MVA EGs are separated into two categories:

Term	Abbreviation / Acronym	Definition
		- EGs that are connected to the distribution system and only generate into the distribution system - EGs that are connected to the distribution system, generate and take load from the system
Energy (or usage)		The amount of electricity consumed by a customer (or all customers) over a period of time. Energy is measured in terms of watt hours (Wh), kilowatt hours (kWh), megawatt hours (MWh) or gigawatt hours (GWh).
Feed-in Tariff	FiT	The rate that is to be paid for the excess energy generated by customers and fed back into the electricity grid under the Queensland Solar Bonus Scheme. The FiT rate is determined by the Queensland Government and is paid by the purchaser of the excess energy.
Fixed charge		A type of charge (charging parameter) included in network tariff structures which is levied on a fixed dollar amount per day.
High Voltage	HV	Refers to the network at 11 kV or above.
Inclining Block Tariff	IBT	A type of network tariff where the price per kWh increases as consumption thresholds are crossed during a particular time period.
Large customer classification		As per tariff class assignment process for customers with consumption greater than 100 MWh per year.
Large customer connection	LCC	Large customer connections are those connections that fall within the tariff classes of Individually Calculated Customer (ICC) and Connection Asset Customer (CAC) including embedded generators with installed capacity greater than or equal to 30 kVA.
Long Run Marginal Cost	LRMC	An estimate of the cost (long term variable investment) of augmenting the existing network to provide sufficient capacity for one additional customer to connect to the network or an additional MW of demand.
Low Voltage	LV	Refers to the sub-11 kV network
Maximum demand		The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period, such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.
Micro Generator		AS4777-compliant generators with an installation size of less than 10 kW (single phase) or 30 kW (three phase) connected to the LV network.
National Electricity Law	NEL	The legislation that establishes the role of the AER as the economic regulator of the NEM and the regulatory framework under which the AER operates.
National Electricity Market	NEM	The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.
National Electricity Rules	NER (the NER)	The legal provisions (enforced by the AER) that regulate the operation of the NEM and the national electricity systems, the activities of market participants and the provision of connection services to retail customers.
National Metering Identifier	NMI	A unique number assigned to each metering installation.
Network capacity		The maximum demand (kW) that the distribution network can provide for at any one time.
Network Coupling Point	NCP	The point at which connection assets join a distribution network, used to identify the distribution service price payable by a customer.
Network Tariff Code	NTC	Ergon Energy's nominated code that represents the network tariff being charged to customers for network services.

Term	Abbreviation / Acronym	Definition
Network Use of System	NUOS	The tariff for use of the distribution and transmission networks. It is the sum of both Distribution Use of System (DUOS) and DPPC.
New customer		New customer means a new connection to the network.
Power factor		Power factor is the ratio of kW to kVA, and is a useful measure of the efficiency in the use of the network infrastructure. The closer the power factor is to one (1), the more efficiently the network assets are utilised. Power factor = kW / kVA
Site-specific charge		This charge is calculated for a site and is specific to the individual connection point.
Small customer classification		As per tariff class assignment process for customers with consumption less than 100 MWh per year.
Smart meter		Digital, interval and advanced Type 1-4 meters. Meters capable of measuring electricity usage in specific time intervals and enabling tariffs that can vary by time of day.
Solar Photovoltaic	Solar PV	A system that uses sunlight to generate electricity for residential use. The system provides power for the premises with any excess production feeding into the electricity grid.
Standard Control Service	SCS	Distribution services that are central to electricity supply and therefore relied on by most (if not all) customers. This service classification includes network services (e.g. construction, maintenance and repair of the network), basic connection services and Type 7 metering services (i.e. unmetered connections such as traffic lights).
Tariff		The set of charges applied to a customer in the respective billing period. A tariff consists of one or more charging parameters that comprise the total tariff rate.
Tariff class		A class of customers for one or more <i>direct control services</i> who are subject to a particular tariff or particular tariffs (as per chapter 10 of the NER).
Tariff Structure Statement	TSS	Document prepared in accordance with Part I of chapter 6 of the NER, setting out Ergon Energy's network price structures and indicative tariffs that will apply over each year of the regulatory control period. Ergon Energy submitted its 2020-25 TSS proposal to the AER in December 2019. Once approved, the TSS takes effect from 1 July 2020.
Threshold demand		The amount by which a SAC Large customer's metered monthly actual kW maximum demand is adjusted for the purposes of calculating the demand component of their network tariff. The actual demand charge for any time demand tariffs and the peak and off-peak demand charges for the STOUT tariffs are applied to the kW amount by which the recorded monthly maximum demand exceeds the relevant threshold. This demand may occur at any time during the month (actual demand charge and off-peak demand charge) or during a set peak period (peak charge). Where the monthly metered maximum demand is less than the demand threshold, the chargeable demand for that month is set to zero.
Time-of-use	ToU	A type of network tariff where the price per kWh varies according to when the consumption occurs. The TOU tariff may apply a different price during peak, shoulder and off-peak periods.
Transmission Use of System charge	TUOS	Superseded terminology for DPPC which are charges incurred for use of the transmission network.
Unmetered supply		A customer who takes supply where no meter is installed at the connection point.

Term	Abbreviation / Acronym	Definition
Usage or Volume charge		A type of charge (charging parameter) included in network tariff structures which is calculated using the customer's metered energy (kWh) consumption. It may be based on a flat rate, an inclining block or TOU charging structure (depending on the customer's applicable network tariff). This part of the tariff seeks to reflect costs not directly allocated to network drivers and costs that are proportional to the size of the customer.
Weekday		Weekdays include government gazetted full day public and bank holidays i.e. State, regional and local public holidays.
Workday		Workdays exclude government gazetted full day public holidays but include bank, regional and local holidays as well as part day gazetted public holidays (e.g. Christmas eve).